

Updated September 30, 2026

Homework problems for AMAT 540A (Topology I), Fall 2026. Over the course of the semester I'll add problems to this list, with each problem's due date specified. Each problem is worth 2 points.

Solutions will be gradually added (and may be hastily written without proofreading).

Problem 1 (due Weds 9/2): Let $f: A \rightarrow B$ be a function. Suppose that for all $A_0 \subseteq A$, $f^{-1}(f(A_0)) = A_0$. Prove that f is injective.

Solution: Let $a, a' \in A$ with $f(a) = f(a')$. Set $A_0 = \{a\}$, so $a' \in f^{-1}(f(A_0))$. By hypothesis $a' \in A_0$, so $a = a'$. $\square$

Problem 2 (due Weds 9/2): Let A be a set and let $p_1: A \times A \rightarrow A$ be the projection function $p_1(a, a') = a$. Let $R \subseteq A \times A$ be a relation such that the image $p_1(R)$ equals all of A . Prove that if R is symmetric and transitive then it is reflexive.

Solution: Suppose R is symmetric and transitive. Let $a \in A$. Since $p_1(R) = A$ there exists $(a_1, a_2) \in R$ with $p_1(a_1, a_2) = a$. By construction of p this means $a_1 = a$, so a is related to a_2 under R . By symmetry a_2 is related to a , and by transitivity a is related to a , so we conclude R is reflexive. $\square$

Problem 3 (due Weds 9/2): Let $\mathcal{A}$ be a collection of sets and let $U = \bigcup_{A \in \mathcal{A}} A$. Prove that if $\mathcal{P}(U) = \bigcup_{A \in \mathcal{A}} \mathcal{P}(A)$ then $U \in \mathcal{A}$.

Solution: Since $U \in \mathcal{P}(U)$, our hypothesis ensures that $U \in \bigcup_{A \in \mathcal{A}} \mathcal{P}(A)$, i.e., $U \in \mathcal{P}(A)$ for some $A \in \mathcal{A}$. This means $U \subseteq A$, and $A \subseteq U$ by definition, so we conclude $U = A \in \mathcal{A}$ as desired. $\square$

No class Monday 9/7 (Labor Day)

Problem 4 (due Weds 9/9): Prove that $\{U \subseteq \mathbb{N} \mid U = \emptyset \text{ or } |\mathbb{N} \setminus U| < 2026\}$ is not a topology on $\mathbb{N}$.

Solution: Let $U = \mathbb{N} \setminus \{1, \dots, 2025\}$ and $V = \mathbb{N} \setminus \{2026\}$, so U, V are both in this set, but $U \cap V = \mathbb{N} \setminus \{1, \dots, 2026\}$ is not in this set. $\square$

Problem 5 (due Weds 9/9): Let X be a set and $\sim$ an equivalence relation. Write $[x]$ for the equivalence class of $x \in X$ with respect to $\sim$. Prove that $\mathcal{B} = \{[x] \mid x \in X\}$ is a basis for a topology on X .

Solution: Since $x \in [x]$ for all x we have the first axiom. Now suppose $x \in [y] \cap [z]$. Then $x \sim y$ and $x \sim z$, so $[y] \cap [z] = [x] \cap [x] = [x]$, so $x \in [x] \subseteq [y] \cap [z]$ reveals the second axiom. $\square$

Problem 6 (due Weds 9/9): Let $\mathcal{T} = \{U \subseteq \mathbb{R} \mid U = \emptyset \text{ or } \mathbb{R} \setminus U \text{ is countable}\}$. Is $\mathcal{T}$ a topology on $\mathbb{R}$? Prove or disprove.

Solution: Yes. First note that $\emptyset \in \mathcal{T}$ by construction, and $\mathbb{R} \in \mathcal{T}$ since $\mathbb{R} \setminus \mathbb{R} = \emptyset$ is countable. Next for any family $U_\alpha \in \mathcal{T}$, without loss of generality none empty, we have that $\mathbb{R} \setminus U_\alpha$ is countable for all α , so $\mathbb{R} \setminus (\bigcup_\alpha U_\alpha)$ is countable (being a subset of any of these), hence $\bigcup_\alpha U_\alpha \in \mathcal{T}$. Finally, for $U, V \in \mathcal{T}$, if one of them is empty their intersection is empty hence in $\mathcal{T}$, and otherwise $\mathbb{R} \setminus U$ and $\mathbb{R} \setminus V$ are countable, hence $\mathbb{R} \setminus (U \cap V) = (\mathbb{R} \setminus U) \cup (\mathbb{R} \setminus V)$ is countable, so $U \cap V \in \mathcal{T}$. $\square$

Problem 7 (due Weds 9/16): Let X be an infinite set with the finite complement topology. Prove that $\mathcal{B} := \{B \subseteq X \mid |X \setminus B| \text{ is a prime number}\}$ is a basis for this topology.

Solution: Let U be open in X and $x \in U$. Since $U \neq \emptyset$, $|X \setminus U| < \infty$. Choose a prime number p greater than $|X \setminus U|$. Since X is infinite so is $U \setminus \{x\}$, hence we can choose $P \subseteq U \setminus \{x\}$ such that $|X \setminus U| + |P| = p$. Now setting $B := U \setminus P$ we get $|X \setminus B| = |X \setminus U| + |P| = p$, so $B \in \mathcal{B}$, and $x \in B \subseteq U$, so $\mathcal{B}$ is a basis. $\square$

Problem 8 (due Weds 9/16): Let X be a set with at least two elements, with a total order $\leq$ such that for all $a < b$ in X there exists $c \in X$ with $a < c < b$. (Here $<$ means “ $\leq$ and $\neq$ ”.) Prove that the order topology on X is not the same as the discrete topology.

Solution: Since X has at least two elements, the other hypothesis ensures that there exists $x \in X$ that is neither a global min nor max. We claim $\{x\}$ is not open in the order topology, hence the order topology is not discrete. Suppose $\{x\}$ is open, so either $\{x\} = (a, b)$ for some $a < b$, or $\{x\} = [a_0, b)$ for some global min a_0 and some b , or $\{x\} = (a, b_0]$ for some a and some global max b_0 . Since $a_0 \in [a_0, b)$ and $b_0 \in (a, b_0]$, but x is not a global min or max, these cases cannot happen, so $x = (a, b)$ for some $a < b$. Hence $a < x < b$, but the hypotheses ensure there exist $c, d \in X$ with $a < c < x$ and $x < d < b$, so (a, b) has more elements than just x in it, a contradiction. $\square$

Problem 9 (due Weds 9/16): Let X be the set from Problem 8, with total order $\leq_X$ having the same hypotheses (so the order topology is not discrete). Let Y be another totally ordered set with order $\leq_Y$ and at least two elements. Let $\leq_{lex}$ be the lexicographic order on $X \times Y$.

Prove that the order topology on $X \times Y$ using $\leq_{lex}$ does not equal the product topology using $\leq_X$ and $\leq_Y$.

Solution: [Let me write $\langle x, y \rangle$ for ordered pairs, to distinguish them from open intervals (a, b) .] Since X is as in Problem 8, we know we can choose some $a < x < b$ in X . Let $y_1 < y_2$ in Y . Consider the lexicographic interval $L = (\langle x, y_1 \rangle, \langle b, y_2 \rangle)$, so L is open in the order topology. Note that L contains the point $\langle x, y_2 \rangle$. Now we claim that no basic open set $U \times V$ in the product topology can satisfy $\langle x, y_2 \rangle \in U \times V \subseteq L$. Indeed, this would mean that $x \in U$ and hence we could choose an interval (x_1, x_2) in X with $x \in (x_1, x_2) \subseteq U$. Now choose x_3 with $x_1 < x_3 < x$, so the point $\langle x_3, y_2 \rangle$ lies in $U \times V$, but $\langle x_3, y_2 \rangle$ is not lexicographically greater than $\langle x, y_1 \rangle$, contradicting that $U \times V \subseteq L$. $\square$

Exam in class on Weds 9/23

Problem 10 (due Weds 9/23): Let X and Y be topological spaces and consider $X \times Y$ with the product topology. Prove that if $C \subseteq X$ and $D \subseteq Y$ are closed then $C \times D$ is closed.

Problem 11 (due Weds 9/23): Let X and Y be infinite sets with the finite complement topology. Prove that the finite complement topology on $X \times Y$ is not a refinement of the product topology on $X \times Y$.

Problem 12 (due Weds 9/23):

Let X be a set with at least two elements, with the *excluded point topology* $\{U \mid \spadesuit \notin U\} \cup \{X\}$, where $\spadesuit$ is some fixed element. (It's pretty obvious this is a topology, you don't need to bother proving that.) Prove that X has exactly two sets that are simultaneously open and dense.

Problem 13 (due Weds 9/30): Let X be a topological space and $S \subseteq Y \subseteq X$. Suppose that S is dense in Y using the subspace topology on Y , and Y is dense in X . Prove that S is dense in X .

Problem 14 (due Weds 9/30): Let X be the topological space from Problem #5. Prove that if X is Hausdorff then it is discrete.

Problem 15 (due Weds 9/30): Let X and Y be topological spaces and let $f: X \rightarrow Y$ be an injective function. Suppose X is Hausdorff and Y has the finite complement topology. Prove that f is continuous.

Problem 16 (due Weds 10/7): Let X and Y be topological spaces and let $y_0 \in Y$. Let $\iota: X \rightarrow X \times Y$ be $\iota(x) := (x, y_0)$, and view $X \times Y$ with the product topology. Prove that ι is continuous.

Problem 17 (due Weds 10/7): Let Λ be an infinite set. Let $(X_\alpha)_{\alpha \in \Lambda}$ be a family of topological spaces and let $X = \prod_{\alpha \in \Lambda} X_\alpha$ be their product, with the product topology. For each $\alpha \in \Lambda$ let $U_\alpha \subseteq X_\alpha$ be open, and suppose that $\emptyset \neq U_\alpha \neq X_\alpha$ holds for all $\alpha \in \Lambda$. From the definition of product topology, we know that $\prod_{\alpha \in \Lambda} U_\alpha$ is not a basic open set. Prove that it is not even an open set at all.

Problem 18 (due Weds 10/7): Let $C = \prod_{\mathbb{N}} \{0, 1\} = \{0, 1\} \times \{0, 1\} \times \cdots$ be the Cantor space. Let C_0 be the basic open set $C_0 := \{0\} \times \{0, 1\} \times \{0, 1\} \times \cdots$. Prove that $C_0 \cong C$. [For fun, you can try to prove that every basic open set is homeomorphic to C (this might have an annoying number of cases though, I'm not sure).]

Problem 19 (due Weds 10/14):

Problem 20 (due Weds 10/14):

Problem 21 (due Weds 10/14):
